

Run C Record

RUN C — BASELINE RESULT AND PROTOCOL DEVIATION NOTE

CIP-BTSB-BT-001 · Supporting Record R-C

CIP — Compute Infrastructure Platform · S. Mohammed Idreesh · Chennai Run date: 26 August 2026 · Issued before Runs B and A were performed

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

To record the baseline run of CIP-BTSB-BT-001, and to state — **before Runs B and A are performed** — one deviation from the pre-registered protocol that Run C’s timing made necessary.

Issuing this before the remaining runs is deliberate. A protocol change decided after seeing all the data is indistinguishable from a change made to suit the data. This one is fixed, dated and published while two thirds of the dataset does not yet exist.

2. RUNS PERFORMED

Two attempts were made. Both are reported.

Run	Outcome	Reason
C-1	ABORTED at 12.4 min	Endpoint unreachable — apparatus limitation, §4
C-2	COMPLETE	Endpoint reached at 22.04 min

2.1 Configuration

Both runs: plates H and C in direct contact through a thin thermal-paste film. **No gasket, no test layer.** Three ceramic cartridge heaters in series, 11.8 V × 1.16 A = 13.7 W electrical (Record G3). Four G-clamps. Heated from above.

The two runs differ **only** in insulation, per §4.

3. RUN C-1 — ABORTED

3.1 Result

Power on	t = 153 s
Power off	t ≈ 897 s
Heating duration	12.4 min
Peak TC1 / TC2	63.75 / 61.50 °C
Baseline offset TC1 – TC2	–0.14 °C (sd 0.63)

3.2 Why it was aborted

A heat-loss model fitted to 700 s of data gave:

$C \cdot dT/dt = P - k \cdot (T - T_{\text{ambient}}), \quad C = 172.5 \text{ J/K}$	
P power entering the plates	= 9.07 W (66% of 13.7 W electrical)
k loss coefficient	= 0.221 W/K
tau time constant	= 13.0 min
→ equilibrium TC2	≈ 76 °C

The model reproduced the measured mean plate temperature to within 0.3 °C across the run. **It showed that TC2 could not reach the pre-registered 85 °C endpoint in that configuration** — the fixture was shedding heat as fast as the heaters supplied it.

The run was terminated at 12.4 min rather than at the §5.5 thirty-minute limit, because the ceiling had already been established from the data.

3.3 What it produced

C-1 is not a discarded run. It is the measurement that produced the fixture’s heat loss coefficient, and it is the reason the apparatus was changed before any usable dataset existed. Its raw CSV is published as `run_C_ABORTED_20260826.csv`.

4. APPARATUS CHANGE BETWEEN C-1 AND C-2

Insulation only. No change to heaters, supply, plates, sensors, clamping positions or firmware.

	C-1	C-2
Top	200 × 100 mm blanket draped over the clamps, air gap beneath	Blanket pressed into direct contact with the heaters
Clamps	bare steel, exposed	wrapped
Edges	bare	packed with blanket
Bottom	175 × 40 mm strip	enlarged

The four G-clamps were the dominant loss path: steel in direct contact with a heated plate, projecting into ambient air, functioning as cooling fins.

Measured effect:

loss coefficient	0.221 → 0.1061 W/K	(52% reduction)
time constant	13.0 → 27.1 min	
TC2 ceiling	~76 → ~126 °C	

5. RUN C-2 — THE BASELINE

5.1 Timing

Power ON	16:40 clock · t = 87.5 s
Power OFF	17:02 clock · t = 1410.0 s
Samples	1,600 · zero sensor faults

5.2 Baseline offset — measured before power

TC1	36.14 °C	TC2	36.24 °C
TC1 – TC2	= –0.098 °C	sd	0.597

No fixed correction is applied (Record G2 §5.1). This value is the per-run baseline for Run C and is recorded as such.

For continuity: the same probe pair read **–0.03 °C** with their tips held together in air during the Gate 2 boiling checks. Mounted on two separate plates in a clamped fixture they read **–0.10 °C** apart.

5.3 PRIMARY ENDPOINT

t85 (TC2 = 85.0 °C)	= 1322.5 s
	= 22.04 minutes

By linear interpolation between the straddling samples, per §6.1.

Peak TC1 88.00 °C at t = 1451 s; peak TC2 86.75 °C at t = 1478 s. Both peaks occur **after** power off, which is the expected thermal lag of the fixture.

5.4 Delivered power — the pre-specified method

§6.3 specifies a linear least-squares fit to the mean of TC1 and TC2 over the interval where that mean lies between 40 and 70 °C.

n	= 676 samples
slope	= 0.04480 °C/s = 2.688 °C/min
P	= slope × 172.5 J/K = 7.73 W

DELIVERED POWER = 7.73 W. This is the governing figure, and it is used for the energy balance in §6.4.

56% of the 13.7 W electrical input reaches the aluminium.

5.5 Delivered power — supplementary, NOT pre-specified

A two-parameter fit separating delivered power from heat loss gives:

P	=	9.49 W	k	=	0.1061 W/K
tau	=	1626 s (27.1 min)	ceiling T_mean	≈	126 °C

This figure is reported for completeness and is NOT used in the analysis. It is higher than the pre-specified figure because the simple linear fit of §6.3 is taken over an interval in which losses are already significant, and therefore understates the power at the start of the run.

The pre-specified method was fixed before the data existed. **It governs. The more flattering number does not.**

5.6 Results against predictions

Quantity	Predicted (§7.2)	Measured	Verdict
TC1 – TC2 at steady state	1 - 3 °C	2.23 °C	✔ within band
t85 Run C	10 - 16 min	22.04 min	⚠ longer
Delivered power	10 - 15 W	7.73 W	✗ below band

On t85. §7.2 states: “Predicted times are lower bounds. Heat loss to insulation, edges and ambient is not modelled, so measured times are expected to exceed these figures.” The measurement behaved as the pre-registration anticipated.

On delivered power. This is a straightforward failed prediction and is reported as such. The predicted band assumed a smaller fraction of input power would be lost before reaching the plates. **56% delivery was not anticipated.** No falsification criterion in §9 depends on delivered power, so the hypotheses are unaffected — but the prediction was wrong and is recorded as wrong.

6. PROTOCOL DEVIATION — TERMINATION TIME

DEVIATION. §5.5 terminates each run at, among other criteria, **30 minutes elapsed**. For Runs B and A this limit is extended to **45 minutes**.

Decided 26 August 2026, after Run C and before Runs B and A were performed.

6.1 Reason

Run C reached its endpoint at 22.04 min against a predicted 10–16 min — a factor of about 1.7. Applying the same factor to the §7.2 predictions:

RUN B	predicted 13-19 min	→	expect ~27 min
RUN A	predicted 17-27 min	→	expect ~37 min

Run A would reach the 30-minute limit before TC2 reached 85 °C, and the primary endpoint would not exist for the run that carries the primary result.

6.2 What is NOT changed

TC1 = 110 °C early stop	UNCHANGED
TC1 = 120 °C hard abort	UNCHANGED
any safety abort criterion	UNCHANGED
primary endpoint t85	UNCHANGED
analysis method §6	UNCHANGED
falsification criteria §9	UNCHANGED
predicted values §7	UNCHANGED

The 30-minute figure was a practicality bound on run length, not a safety limit. The temperature limits on TC1 are the safety limits and they are untouched. Whichever criterion is met first still terminates the run.

6.3 Why it is being stated now

Because the honest moment to change a protocol is **before** the data it affects exists. Two thirds of this dataset — Runs B and A — has not been collected. This document is published with its date so that the sequence can be verified by anyone.

7. STATUS

[✓]	RUN C	complete	t85 = 22.04 min
[]	RUN B	mineral oil, 25.6 mL	
[]	RUN A	paraffin wax, 23.0 g	

Raw data published as `run_C2_20260826.csv` and `run_C_ABORTED_20260826.csv`, both unedited.